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Geiger-Müller-Counter

13606-99

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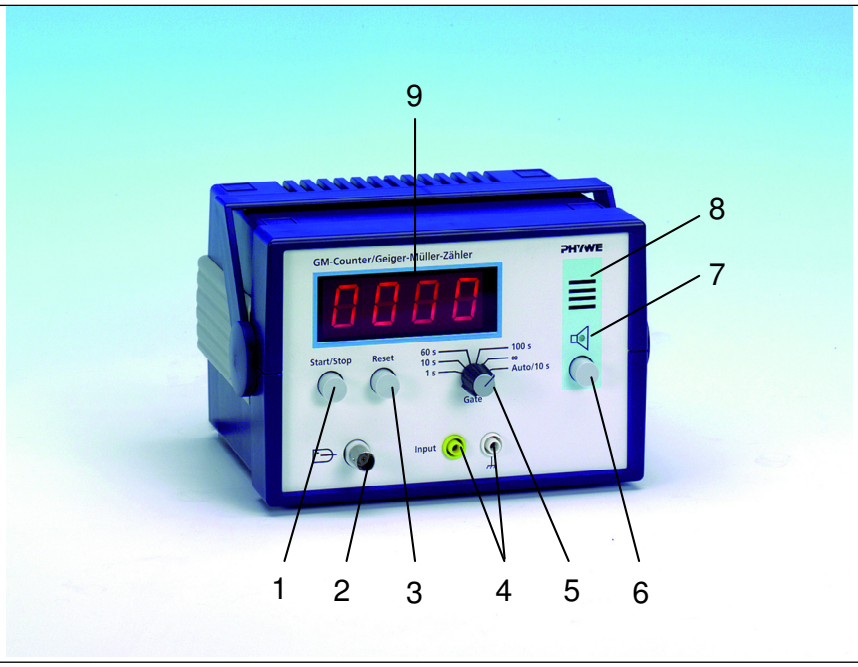

Fig. 1: 13606-99 Geiger-Müller-Counter

Operating instructions

The unit complies with the corresponding EC guidelines.

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1 SAFETY PRECAUTIONS

Caution!

- Carefully read these operating instructions completely before operating this instrument. This is necessary to avoid damage to it, as well as for user-safety.
- Only use the instrument for the purpose for which it was designed.
- Check that your mains supply voltage corresponds to that given on the type plate fixed to the instrument.
- Install the instrument so that the on/off switch and the mains connecting plug are easily accessible.
- Do not cover the ventilation slots.
- Take care that no liquids or objects enter in through the ventilation slots.
- Only use the instrument in dry rooms in which there is no risk of explosion.
- Do not start up this instrument in case of visible signs of damage to it or to the line cord.

2 PURPOSE AND CHARACTERISTICS

This instrument serves primarily as a Geiger-Müller counter, for which purpose a counter tube must be connected to it.
It has, in addition to the counter tube input, a TTL input which enables it to be used to measure frequencies, e.g. in combination with sound probe 03542-00. The Geiger-Müller counter is designed both for student experiments and for practical work in Physics, but is also excellently suited for demonstration experiments because of the large digital display.

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3 FUNCTIONAL AND OPERATING ELEMENTS

The connecting cable supplied enables the instrument to be connected to the AC mains. It plugs into the connecting plug unit in the back of the instrument. A fuse holder in the lower part of the connecting plug unit can only be opened after the plug is removed, e.g. with a screwdriver. For replacement 5 mm x 20 mm fine fuses, see the type plate.
The main switch for turning the instrument on and off is also in the immediate vicinity of the connecting plug unit at the back of the housing. All other parts and operating elements are situated on the front of the instrument (see the Figure):

1 *START/STOP button*
for starting a measurement. When this button is pressed before the end of the selected measuring time, the counting procedure is stopped and the count displayed.
In this case, a new measurement cannot be started until the display is cleared with the RESET button.

2 *Counter input*
BNC socket for connection of a counter tube with a BNC cable. The socket carries the direct voltage required for the operation of the counter tube.

3 *RESET button*
for returning the display to 0000. When this button has been pressed, the counter can be restarted with button 1.

4 *INPUT pair of sockets*
4 mm safety sockets for feeding in a TTL signal voltage for impulse counting.

5 *Rotary switch*
for selecting the time of measurement or operating mode. There is an option of the following three modes of operation:

Individual measurements with fixed time of measurement:
The switch positions „1 s“, „10 s“, „60 s“ and „100 s“ are available here. After starting the counting procedure with the START/STOP button, the measurement stops on completion of the selected time of measurement.
If the display is not cleared with the RESET button before a new measurement is started, the impulses counted are added to those in the existing display.
A premature termination of a measurement in progress is possible at any time by pressing the START/STOP button. Following this, the START/STOP button is without function and is only freed after the display has been cleared with RESET.
Individual measurements with manual stoppage of the counting procedure:
In the switch position „∞“, after starting a counting procedure with the START/STOP button, this keeps running until it is stopped by pressing the START/STOP button again. The measurement can be started up again at any time by pressing button 1; the impulses counted are added to those already displayed.
As soon as the count displayed exceeds 9999 (overflow), the measurement is stopped and 9999 is shown in the display.

Automatic succession of measurements:
In the switch position „Auto/10 s“, after starting with button 1, a counting procedure runs for 10 seconds. The count measured remains on display. while in the background a new counting procedure, starting from zero, runs. When the time of measurement is completed, the new impulse count appears in the display. In this way, a new measured value is obtained at exactly 10 second intervals.
This mode of operation is particularly suitable for the comfortable recording of decay curves (half-life determination) of short-lived radionuclides. To stop the measurement, press the START/STOP button.

6 *Loudspeaker button*
for turning the loudspeaker on and off. The switching condition of the loudspeaker is shown by the LED 7.

7 *Light emitting diode*
lights up, when the loudspeaker is on.

8 *Loudspeaker*
for the acoustic reporting of the impulses. The impulses can be heard even when the counting procedure has not been started.

9 *Display*
for reporting the result of the count.

4 HANDLING

Connect the Geiger-Müller counter to the AC mains. As signal source, connect either a suitable counter tube (see „List of equipment“) to input 2 with a screened-off cable, or a TTL signal source to the input socket 4. The connecting cable must not be longer than 3 m. The two inputs cannot be used simultaneously.

Switch the instrument on with the main switch at the back of the housing. Set switch 5 to the required time of measurement or mode of operation and start measurement by pressing button 1. Further possibilities and details on the functioning of the instrument are given in the section „Parts and operating elements“.
Do not press button 1 while a measurement with preselected time of measurement is running; the count displayed would be of no use as a result, as the counting time is not known. This function is only meant for a quick abortion in faulty situations. When the counter does not start by additive counts when button 1 is pressed, it is certainly because the last measurement was prematurely broken off. In this case, the counter is only ready to operate after the RESET button has been pressed.

5 NOTES ON OPERATION

This instrument is only to be put into operation under specialist supervision in a controlled electromagnetic environment in research, educational and training facilities (schools, universities, institutes and laboratories).
This means that in such an environment, no mobile phones etc. are to be used in the immediate vicinity. The individual connecting leads are each not to be longer than 2 m.

The instrument can be so influenced by electrostatic charges and other electromagnetic phenomena that it no longer functions within the given technical specifications. The following measures reduce or do away with disturbances:
Avoid fitted carpets; ensure potential equalization; carry out experiments on a conductive, earthed surface, use screened cables, do not operate high-frequency emitters (radios, mobile phones) in the immediate vicinity.

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6 TECHNICAL DATA

Inputs	Counter tube (BNC socket) TTL (2 x 4 mm safety sockets)
Counter tube voltage	500 V at BNC socket
Digital display	4 digit 7 segment
Height of digits	19 mm
Counting times (individual measurement)	1, 10, 60, 100 s
Counting time (automatic succession of measurements)	10 s, output each 10 s
Other counting times	Manual stop and start function
Power supply	100 to 240 V~, 50 to 560 Hz
Power consumption	1 VA
Housing dimensions (mm)	194 x 140 x 130

7 LIST OF EQUIPMENT

The following counter tubes can be used in conjunction with the Geiger-Müller counter:

Geiger-Mueller Counter tube, type A	09025-11
Geiger-Mueller Counter tube, type B	09005-00
Geiger-Mueller Counter tube, 45 mm	09007-00

8 NOTES ON THE GUARANTEE

We guarantee the instrument supplied by us for a period of 24 months within the EU, or for 12 months outside of the EU. Excepted from the guarantee are damages that result from disregarding the Operating Instructions, from improper handling of the instrument or from natural wear.

The manufacturer can only be held responsible for the function and technical safety characteristics of the instrument, when maintenance, repairs and alterations to the instrument are only carried out by the manufacturer or by personnel who have been explicitly authorized by him to do so.

9 WASTE DISPOSAL

The packaging consists predominately of environmentally compatible materials that can be passed on for disposal by the local recycling service.

Should you no longer require this product, do not dispose of it with the household refuse. Please return it to the address below for proper waste disposal.

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